

The diagonal Amos-type family at real order: a machine-checked quantitative crossing classification

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Abstract

For the one-parameter family of Amos-type expressions

$$B_{\nu,c}(x) = \frac{x}{\nu + c + \sqrt{(\nu + c)^2 + x^2}}, \quad c \in \mathbb{R},$$

whose member $c = \frac{1}{2}$ is the classical Amos-type upper bound for the modified Bessel ratio $\rho_\nu = I_{\nu+1}/I_\nu$, we formalize in Lean 4, at every real order $\nu \geq 0$ over the Γ -power series, the classification of the parameter: $B_{\nu,c}$ is a uniform upper bound for ρ_ν (all $\nu \geq 0$, $x > 0$) exactly when $c \leq \frac{1}{2}$, and a uniform lower bound for every $c \geq 1$, with explicit counterexample witnesses. The classification statement itself is equivalent to known results of Ruiz-Antolín and Segura; for it we claim only the machine-checking. The contribution is the regime between the ends: for every $\nu \geq 0$ and every $c \in (\frac{1}{2}, 1)$ we prove that the fixed diagonal member $B_{\nu,c}$ crosses ρ_ν *exactly once* on $(0, \infty)$ — a transversal crossing located in the explicit window $[\sqrt{1-c}, 2\hat{D}/(2c-1)]$ with $\hat{D} = (\nu + \frac{3}{2})^2 - (\nu + c)^2$, obeying the strict threshold $x_* > x_\dagger = 2(\nu + c)\sqrt{c(1-c)}/(2c-1)$, with globally determined sign on both sides and the two-sided scale law $2(\nu + c)\sqrt{c(1-c)} \leq (2c-1)x_* \leq 2\hat{D}$. Degenerate (critical) contact is *excluded by theorem*, through the exact identity $(\nu + c)^2 D''(x_\dagger) = (2c-1)^3 B_{\nu,c}(x_\dagger)$ for the gap $D = \rho_\nu - B_{\nu,c}$ at any hypothetical critical contact. The entire chain carries the axiom oracle `[propext, Classical.choice, Quot.sound]` and no analytic hypothesis beyond $\nu \geq 0$, $x > 0$; a pre-registered certified interval-arithmetic companion verifies the crossing phenomenon independently of the C5 crossing theorems at 30 parameter pairs spanning the hard regimes, all passing at 128 bits.

1 Introduction

The ratio $\rho_\nu(x) = I_{\nu+1}(x)/I_\nu(x)$ of modified Bessel functions of the first kind admits algebraically calibrated bounds of the shape $x/(\lambda + \sqrt{\mu^2 + x^2})$; the literature on such bounds — Amos [1], Segura [2, 5, 6], Hornik–Grün [3, 7], Ruiz-Antolín–Segura [4] — is rich, and the one-parameter *diagonal* family studied here,

$$B_{\nu,c}(x) = \frac{x}{\nu + c + \sqrt{(\nu + c)^2 + x^2}},$$

*This manuscript and parts of the formal development were produced with AI assistance under the repository’s audit protocol. Every theorem below is machine-checked in Lean 4 against a pinned Mathlib; the author-produced numerical companion of Section 6 is a committed interval-arithmetic transcript; remaining numerics are labelled paper-level where they occur. Repository: <https://github.com/lluiseriksson/THE-ERIKSSON-PROGRAMME>.

is, up to reparametrization, the family whose sharp ends [4] identifies: under $\mu = \nu + 1$, $\alpha = 2c - 1$, the family b_α of [4] is exactly $B_{\nu,c}$. *No novelty is claimed for the family, nor for the qualitative observation that its non-bounding members must cross the ratio* — [4] already notes that for parameters between the sharp ends the graphs cross. Three companion papers of this series formalized the consequence calculus of the $c = \frac{1}{2}$ bound at integer order [11], the bound itself at integer order [12], and the bound at every real order $\nu \geq 0$ over the Γ -power series [13]; the present paper builds directly on the last.

The contribution here is twofold, and the two halves carry different claims:

1. **The classification, machine-checked.** For every real $\nu \geq 0$ and $x > 0$: $\rho_\nu < B_{\nu,c}$ holds uniformly iff $c \leq \frac{1}{2}$ (Theorem 3.1), and for $c \geq 1$ the reverse inequality holds uniformly (Theorem 3.2). As mathematics this is equivalent to results of [4]; the claim is the formalization — uniform in real order, from the series, with explicit witnesses, inside one pinned axiom-clean development.
2. **The quantitative crossing theorem.** For $\nu \geq 0$ and $\frac{1}{2} < c < 1$, the fixed diagonal member $B_{\nu,c}$ crosses ρ_ν exactly once; the crossing is transversal, satisfies an explicit strict threshold, determines the sign of $\rho_\nu - B_{\nu,c}$ globally on both sides, lies in a constructive finite window, and obeys an explicit two-sided scale law (Theorem 3.3). Degenerate contact is excluded by an exact second-derivative identity (Theorem 3.4). We are not aware of a previous theorem giving this complete quantitative crossing classification for the diagonal family; Section 7 compares against the closest antecedents parameter by parameter.

Statuses: **exact** means a Lean theorem in the pinned central repository with clean axiom oracle; **certified** means a committed interval-arithmetic transcript; **paper-level** means ordinary prose, always flagged; **verified** marks an ordinary high-precision computation used only as a cross-check of a certified result, never as evidence. The repository’s largest namespace (`YangMills/`) reflects its original motivation in lattice gauge theory; no reduction to any continuum or gauge-theoretic problem is claimed or used.

2 Setting

All objects live in module `AmosClosure` of the pinned central repository, over the real-order interface of [13]: $I_\nu = \text{besselIReal } \nu$ is the Γ -power series $\sum_{k \geq 0} (x/2)^{\nu+2k} / (k! \Gamma(\nu + k + 1))$ (real exponents via `rpow`), with positivity, the three-term recurrence, termwise differentiability, the Riccati equation

$$\rho'_\nu(x) = Q_x(\rho_\nu(x)), \quad Q_x(y) := 1 - \frac{2\nu + 1}{x} y - y^2, \quad (1)$$

the barrier $\psi_\nu > 2\nu + 1$ for $\psi_\nu(x) = x(1/\rho_\nu - \rho_\nu)$, and the $c = \frac{1}{2}$ bound $\rho_\nu < B_{\nu,1/2}$ (`amosBoundReal_holds`). New objects of this paper: the family $B_{\nu,c} = \text{amosFamily } \nu \ c$, the lower bound $L_\nu = \text{amosLower } \nu$ of Section 4.2, and the crossing gap

$$D(x) = \rho_\nu(x) - B_{\nu,c}(x).$$

Throughout, $a = \nu + c$ and $s = \sqrt{a^2 + x^2}$. Everything is stated at $x > 0$, the `rpow` domain of the development.

3 Main results

Theorem 3.1 (classification of the upper-bound range; `amosFamily_uniform_upper_iff`). *For every $c \in \mathbb{R}$:*

$$\left(\forall \nu \geq 0, \forall x > 0 : \rho_\nu(x) < B_{\nu,c}(x) \right) \iff c \leq \frac{1}{2}.$$

The failure witness for $c > \frac{1}{2}$ is an explicit rational function of c : $x_0(c) = 2/(2c - 1)$ at $\nu = 0$ (`amosFamily_lt_amosLower_witness`).

Theorem 3.2 (the uniform lower half; `amosFamily_lower_of_one_le`). *For every $c \geq 1$, every $\nu \geq 0$ and every $x > 0$, $B_{\nu,c}(x) < \rho_\nu(x)$.*

Theorems 3.1 and 3.2 are machine-checked forms of known mathematics [4]; between them lies the regime $\frac{1}{2} < c < 1$, where $B_{\nu,c}$ bounds on neither side. There — with the strictness of the threshold in item (ii), and through it the unconditional uniqueness and orientation, delivered by Theorem 3.4 below:

Theorem 3.3 (the global crossing theorem; `amosFamily_global_crossing`, `amosFamily_crossing_orientation`). *Fix $\nu \geq 0$ and $\frac{1}{2} < c < 1$. There is exactly one $x_* > 0$ with $\rho_\nu(x_*) = B_{\nu,c}(x_*)$, and:*

- (i) (window) $\sqrt{1-c} \leq x_* \leq \frac{2\hat{D}}{2c-1}$, where $\hat{D} = (\nu + \frac{3}{2})^2 - (\nu + c)^2$ (`amosFamily_crossing_exists`);
- (ii) (strict threshold) $x_* > x_\dagger := \frac{2(\nu+c)\sqrt{c(1-c)}}{2c-1}$ (`crossing_threshold_strict`);
- (iii) (transversality) at the crossing, $D'(x_*) = \frac{B_{\nu,c}(x_*)}{x_*} \left(2c - 1 - \frac{\nu+c}{\sqrt{(\nu+c)^2 + x_*^2}} \right) > 0$ (`crossing_hasDerivAt`, whose factor is positive by `crossing_threshold_strict`);
- (iv) (orientation) $\rho_\nu < B_{\nu,c}$ on $(0, x_*)$ and $\rho_\nu > B_{\nu,c}$ on (x_*, ∞) (`amosFamily_crossing_orientation`); conversely — an immediate trichotomy consequence — the sign of D locates x relative to x_* ;
- (v) (scale law) $2(\nu+c)\sqrt{c(1-c)} \leq (2c-1)x_* \leq 2\hat{D}$ (`crossing_scale`).

Theorem 3.4 (no degenerate contact; `crossing_no_critical_contact`). *Fix $\nu \geq 0$ and $\frac{1}{2} < c < 1$. No crossing of ρ_ν and $B_{\nu,c}$ is a critical point of D : at any hypothetical contact with $D = D' = 0$ — which the slope identity forces to lie exactly at $x_\dagger$ — the second derivative obeys the exact identity*

$$(\nu+c)^2 D''(x_\dagger) = (2c-1)^3 B_{\nu,c}(x_\dagger) > 0,$$

and a critical contact with positive curvature contradicts the below-threshold orientation. Consequently the threshold in Theorem 3.3 is strict and uniqueness and orientation hold unconditionally.

The displayed curvature calculation is derived internally, under the hypothetical contact assumptions, in the proof of `crossing_no_critical_contact`. That declaration exports their contradiction (`False`); it does not export the second-derivative identity as a separate named theorem.

Theorem 3.5 (the Turán inequality at real order; `besselIReal_turan`). *For every $\nu \geq 0$ and $x > 0$, $I_{\nu+1}(x)^2 > I_\nu(x) I_{\nu+2}(x)$; equivalently $\psi_\nu(x) < 2\nu + 2$, completing the sandwich $2\nu + 1 < \psi_\nu(x) < 2\nu + 2$ whose lower half is the barrier of [13]. This inequality is classical (Turán-type inequalities for I_ν go back to Thiruvengatachar–Nanjundiah; see Baricz–Ponnusamy [9]); it is proved here at real order over the Γ -series and used as supporting structure.*

4 The proof

The formal proofs are the authority; this section presents the mathematics in full, at the level of the checklist a referee needs.

4.1 Family algebra: calibration and the nullcline trichotomy

Since $s = \sqrt{a^2 + x^2} > |a| \geq -a$, the denominator $a + s$ is positive for *every* real c , and the conjugate form $B_{\nu,c} = (s - a)/x$ follows from $(s - a)(s + a) = x^2$ (`amosFamily_pos`). The general calibration identity

$$\frac{1}{B_{\nu,c}(x)} - B_{\nu,c}(x) = \frac{2(\nu + c)}{x} \quad (\text{amosFamily_calibration}) \quad (2)$$

extends the $c = \frac{1}{2}$ calibration of [13]. Substituting into the Riccati quadratic of (1): $1 - B^2 = (2a/x)B$, so

$$Q_x(B_{\nu,c}(x)) = \frac{2a - (2\nu + 1)}{x} B_{\nu,c}(x) = \frac{2c - 1}{x} B_{\nu,c}(x) \quad (\text{riccatiQReal_amosFamily}), \quad (3)$$

with the sign trichotomy: zero iff $c = \frac{1}{2}$ (the member *on* the nullcline — exactly the calibration that drives the proof of the $c = \frac{1}{2}$ bound), positive iff $c > \frac{1}{2}$, negative iff $c < \frac{1}{2}$. Since $a \mapsto x/(a + \sqrt{a^2 + x^2})$ is decreasing (`amosFamily_anti`; the non-strict form is all the chain uses), $c \leq \frac{1}{2}$ gives $B_{\nu,c} \geq B_{\nu,1/2} > \rho_\nu$ by [13]: the sufficiency half of Theorem 3.1 (`amosFamily_upper_of_le_half`).

4.2 The lower bound by recurrence inversion

The ratio recurrence $1/\rho_\nu = 2(\nu + 1)/x + \rho_{\nu+1}$ (`besselIReal_ratio_recurrence`) turns the $c = \frac{1}{2}$ upper bound at order $\nu + 1$ into a lower bound at order ν : with $b = \nu + \frac{3}{2}$ and $s' = \sqrt{b^2 + x^2}$,

$$\frac{1}{\rho_\nu(x)} < \frac{2(\nu + 1)}{x} + \frac{s' - b}{x} = \frac{\nu + \frac{1}{2} + s'}{x}, \quad \text{i.e.} \quad \rho_\nu(x) > \frac{x}{\nu + \frac{1}{2} + \sqrt{(\nu + \frac{3}{2})^2 + x^2}} =: L_\nu(x),$$

where the middle step is the algebraic collapse $x^2/(b + s') = s' - b$ (`besselLowerReal_holds`). This is the sharp-at-infinity lower bound of [4]; here it is a three-line corollary of the real-order upper bound, which is the point of recording it. L_ν matches ρ_ν to first order at infinity ($L_\nu \sim 1 - (\nu + \frac{1}{2})/x$), while $B_{\nu,c} \sim 1 - (\nu + c)/x$ (paper-level asymptotics, not used in the proofs); for $c > \frac{1}{2}$ the family member falls below L_ν at explicit finite x : at $\nu = 0$ and $x_0(c) = 2/(2c - 1)$ the inequality $B_{0,c}(x_0) < L_0(x_0)$ reduces, after clearing the calibrated denominators with $(2c - 1)x_0 = 2$, to the rational inequality $\frac{9}{4} - c^2 < 2$ for $c \in (\frac{1}{2}, 1]$ and its routine extension beyond (`amosFamily_lt_amosLower_witness`); chaining $B_{0,c}(x_0) < L_0(x_0) < \rho_0(x_0)$ kills uniformity.

That is the necessity half of Theorem 3.1. For Theorem 3.2: $c \geq 1$ gives $B_{\nu,c} \leq B_{\nu,1} < L_\nu < \rho_\nu$, where $B_{\nu,1} < L_\nu$ is the root comparison $\sqrt{(\nu + \frac{3}{2})^2 + x^2} < \frac{1}{2} + \sqrt{(\nu + 1)^2 + x^2}$, i.e. $\nu + 1 < \sqrt{(\nu + 1)^2 + x^2}$ after squaring (`amosFamily_one_lt_amosLower`).

4.3 The Turán sandwich

$\rho_\nu > B_{\nu,1}$ (from Section 4.2) plus the calibration (2) at $c = 1$ gives $\psi_\nu < 2(\nu + 1)$ (`besselPsiReal_lt`); together with the barrier $\psi_\nu > 2\nu + 1$ of [13] this is the sandwich $2\nu + 1 < \psi_\nu < 2\nu + 2$. The upper half converts, through the recurrence, into $\rho_\nu > \rho_{\nu+1}$ and thence into the Turán inequality of Theorem 3.5 (`besselIReal_turan`). We emphasize provenance: the mathematical content of the sandwich upper half is classical [9]; the formal statement at real order over the Γ -series is what is new here, and the crossing results below do not depend on it.

4.4 Existence: the explicit window

Fix $\frac{1}{2} < c < 1$. At the left end, the product-form upper bound $\rho_\nu(x) < x/(2(\nu + 1)(1 - q_\nu))$ with $q_\nu = (x/2)^2/(\nu + 2)$, valid for $q_\nu < 1$ (`besselRatioReal_lt_product`, from the geometric-domination toolkit of [13]) beats the family member below $\sqrt{1 - c}$: using $\sqrt{a^2 + x^2} \leq a + x^2/(2a)$ one checks $D(x) < 0$ for all $0 < x \leq \sqrt{1 - c}$, uniformly in ν (`crossing_witness_small_le`). At the right end, at $X = 2\hat{D}/(2c - 1)$ with $\hat{D} = (\nu + \frac{3}{2})^2 - (\nu + c)^2$, the two calibrated roots separate by less than $(2c - 1)/4$ and $L_\nu(X) > B_{\nu,c}(X)$, so $D(X) > 0$ (`crossing_witness_large`). Continuity of D (from differentiability) and the intermediate value theorem give a crossing in the window (`amosFamily_crossing_exists`).

4.5 Slope identity, threshold, uniqueness, orientation

The family derivative is $B'_{\nu,c}(x) = a/(s(a + s))$ (`amosFamily_hasDerivAt`). At any zero of D , ρ_ν takes the family value, so by (1) and (3) $\rho'_\nu = \frac{2c-1}{x}B_{\nu,c}$ there, while $a/(s(a + s)) = aB_{\nu,c}/(sx)$ by the conjugate form; hence the *slope identity*

$$D'(\text{zero}) = \frac{B_{\nu,c}}{x} \left(2c - 1 - \frac{a}{s} \right) \quad (\text{crossing_hasDerivAt}). \quad (4)$$

The factor $2c - 1 - a/s$ is negative below, zero at, and positive above $x_\dagger = 2a\sqrt{c(1 - c)}/(2c - 1)$ (where $s = a/(2c - 1)$). Three consequences, in order: every crossing satisfies $x \geq x_\dagger$, since a crossing strictly below the threshold would be a strictly *downward* crossing yet $D < 0$ must persist from the left window endpoint (`crossing_threshold`); above the threshold every crossing is strictly upward (`crossing_pos_right`), and two strictly upward zeros of a differentiable function cannot occur without an intermediate downward zero — excluded — so the crossing is unique (`crossing_unique`); and the sign of D is then globally determined on both sides (`crossing_orientation_after`, `crossing_orientation_below`).

4.6 Exclusion of the degenerate contact

One case survives Section 4.5: a contact exactly at $x_\dagger$, where the slope factor vanishes identically and the crossing would be a critical point of D . At such a contact every quantity is an explicit

algebraic function of (ν, c) : writing $t = 2c - 1 \in (0, 1)$,

$$s = \frac{a}{t}, \quad \rho_\nu = B_{\nu,c} = \sqrt{\frac{1-c}{c}}, \quad B_{\nu,c}x = s - a = \frac{a(1-t)}{t}, \quad x^2 = \frac{a^2(1-t)(1+t)}{t^2},$$

and $2\nu + 1 = 2a - t$. Differentiating the Riccati equation along the flow and the family derivative once more (`crossing_hasDerivAt_everywhere` provides D 's derivative as an explicit function everywhere; that function is differentiated once more inside the tangency proof), the two curvatures collapse to

$$\rho''_\nu = \frac{B}{x^2}[(2\nu+1)(1-t) - 2tBx] = -\frac{t^3 B}{a^2(1+t)}, \quad B'' = -\frac{a(x/s)(a+2s)}{s^2(a+s)^2} = -\frac{t^3 B(t+2)}{a^2(t+1)},$$

whence the exact identity of Theorem 3.4: $D'' = t^3 B/a^2$, i.e. $(\nu + c)^2 D'' = (2c - 1)^3 B_{\nu,c} > 0$. In the formal development the identity is proved division-free in exactly this form; positivity needs only $2c - 1 > 0$, $\nu + c > 0$, $B_{\nu,c} > 0$. The contradiction: at a critical contact, $D' = 0$ and $D'' > 0$ force $D' < 0$ on a left interval, so D is strictly decreasing into the contact and $D(y) > D(x_\dagger) = 0$ just left of it; but $y < x_\dagger$ is strictly below the threshold, where the orientation of Section 4.5 gives $D(y) < 0$ (`crossing_no_critical_contact`). No critical contact exists; the threshold is strict (`crossing_threshold_strict`); uniqueness and orientation are unconditional (`crossing_unique_unconditional`); and the packaged existence-and-uniqueness statement is `amosFamily_global_crossing`. The scale law of Theorem 3.3(v) is then immediate from the threshold and the window (`crossing_scale`).

5 Scope

All results are for the in-core Γ -power-series definition of I_ν at real order $\nu \geq 0$, $x > 0$; no identification with an external special-functions library object is claimed (none exists at the pinned Mathlib to identify with), though the identification theorem of [13] ties the object to the factorial series at every integer order. Theorems 3.1 and 3.2 are machine-checked forms of known mathematics [4], and the crossing *phenomenon* for intermediate parameters is likewise observed there; the claims of this paper are (i) the formalization, uniform in real order, and (ii) the quantitative crossing package of Theorems 3.3 and 3.4: unique transversal crossing, strict explicit threshold, bilateral orientation, constructive window, two-sided scale law, degenerate contact excluded. We are not aware of a previous theorem giving this complete quantitative crossing classification for the diagonal family.

6 The certified companion

A certified interval-arithmetic companion verifies the crossing phenomenon *independently of the C5 crossing theorems* — without consulting any C5 crossing, uniqueness, orientation, or scale-law theorem: its protocol — registered in the development's charter in its own commits, strictly before the certified run and before any certified result was observed — mandates that verdicts be decided *only* by ball signs of D and D' , never by consulting the crossing theorems; that the bisection search interval come from the fixed rule $[(1-c)/4, 8(\nu+2)^2/(2c-1)]$, deliberately *wider* than the Lean window, with the theoretical window checked afterwards rather than searched; and that failures escalate a precision ladder (128 to 4096 bits) rather than widen intervals. The

grid of 30 pairs, $\nu \in \{0, \frac{1}{2}, 1, \pi, 10, 100\} \times c \in \{0.501, 0.6, 0.75, 0.9, 0.999\}$ (π a certified ball), forces the hard regimes: $c \downarrow \frac{1}{2}$ (the crossing escapes to infinity — at $(\nu, c) = (100, 0.501)$ the search interval reaches 4.2×10^7), $c \uparrow 1$ (the crossing value $B(x_+) \rightarrow 0$), $\nu = 0$, large order, and the doubly-extreme corners. Per pair, four certificates, all by ball sign: a bracketing box $X = [\text{lo}, \text{hi}]$ with certified opposite signs of D at its ends (certified bisection; midpoints are outward-rounded balls — for non-dyadic parameters the registered exact-dyadic midpoints are not representable — a strictly conservative refinement, since wider endpoint balls make every certificate harder, never easier); transversality $\inf_{x \in X} D'(x) > 0$ by interval evaluation over the hull; window inclusion $X \subset [x_+, x_+]$; and both sides of the scale law. The ratio is evaluated by the backward recurrence $\rho_\alpha = x/(2(\alpha+1) + x\rho_{\alpha+1})$ seeded at order $\nu + N$ with the *interval* $[0, 1]$ — valid by positivity and the $c = \frac{1}{2}$ bound of [13], both outside the forbidden theorem set — so no exponentially large value of I_ν is ever formed; the seed contraction is certified a posteriori by the final ball width, and a too-wide ball can only produce an undecided verdict, never a wrong one. All 30 pairs pass all four certificates at the first ladder rung (128 bits); an independent ordinary high-precision pass (labelled *verified*, not certified) agrees at all 30. Two illustrative rows: at $(100, 0.501)$ the certified box $[50250.8995 \dots]$ (width $\sim 3.5 \times 10^{-14}$) clears the threshold $x_+ = 50250.3995 \dots$ with certified D' ball $= 3.95 \times 10^{-13} \pm 4.1 \times 10^{-16} > 0$; at $(\frac{1}{2}, 0.501)$ the box sits at $x_* = 501.00000000000000 \dots$, reflecting the closed (hyperbolic-cotangent) form of $\rho_{1/2}$: by the calibration (2), the crossing equation is $\psi_{1/2}(x_*) = 2(\nu + c)$, and $\psi_{1/2}(x) = 2 + 1/(x - 1)$ up to exponentially small error, so $x_* \approx 1 + \frac{1}{2c-1} = \frac{2c}{2c-1} = 501$ — a paper-level remark; the certified box needs no such structure. The companion is *not* part of the proof, and the proof does not cite it.

7 Related work

The comparison below follows the development’s pre-registered collation; the antecedents are compared claim by claim.

Ruiz-Antolín–Segura [4] is the exact antecedent of the *family* and of *qualitative existence*: under $\mu = \nu + 1$, $\alpha = 2c - 1$, their b_α family is $B_{\nu,c}$; they identify the sharp ends ($\alpha \leq 0$, i.e. $c \leq \frac{1}{2}$, at $x \rightarrow \infty$; $\alpha \geq 1$, i.e. $c \geq 1$, at $x \rightarrow 0$) and observe that for intermediate parameters the graphs must cross (asymptotic end signs). We declare expressly: the family and the mere existence of some crossing are prior art. [4] contains no uniqueness, orientation, threshold, tangency exclusion, or quantitative localization for the intermediate members.

Hornik–Grün [3] is the closest antecedent in spirit: it characterizes which two-parameter Amos-type expressions $G_{\alpha,\beta}$ are upper or lower bounds, and describes minimal upper bounds as those tangent to the ratio at a single point. The statements are about the *boundary of the bounding region* in the (α, β) -plane; classifying the fixed diagonal $G_{\alpha,\alpha}$ -type member $B_{\nu,c}$ — which for $c \in (\frac{1}{2}, 1)$ is *not* a bound at all — is a different question. Parameter by parameter: [3] yields, for the diagonal, membership/non-membership in the bounding classes; it does not state, for non-bounding diagonal members, the unique simple crossing, its transversality, the explicit threshold x_+ , bilateral orientation, or a scale law.

Segura [6] studies bounds $(\alpha + \sqrt{\beta^2 + \gamma^2 x^2})/x$ -type with coefficients *optimized as functions of a prescribed tangency point*: there the bounding function is a global bound whose graph is tangent to the ratio (osculatory contact by construction). Here the family member is *fixed in advance*, is not a bound, the meeting is transversal, and tangential contact is *excluded by theorem*. Close in machinery, not equivalent in statement.

Hornik–Grün [7] proves single-sign-change results *between two bounds* within their generalized family (bound-versus-bound comparisons); the present single-crossing statement is *ratio-versus-family-member*, a different pair of curves — we flag the terminological near-collision explicitly.

Segura [5] (Riccati/nullcline monotonicity methods) and **Salazar** [8] (Bernstein-positivity certificates for Riccati reductions) supply methodology adjacent to ours; neither presents the quantitative diagonal crossing package.

On the formalization side, the companion papers [11, 12, 13] formalized the $c = \frac{1}{2}$ bound and its consumers; we know of no other machine-checked development of Amos-type bounds or of any crossing classification. The recurrence and derivative identities are [10, 10.29]. The development builds on Mathlib [14].

8 Reproducibility

Toolchain: Lean `leanprover/lean4:v4.29.0-rc6`; Mathlib pinned to commit

`07642720480157414db592fa85b626dafb71355b`

(lakefile and manifest agree); `tectonic 0.15.0`; `python-flint 0.9.0` for the certified companion.

Recorded build facts: at the final mathematics commit

`f1c87fef` (*short id; the release manifest records the full hashes*)

`lake build AmosClosure` completed successfully (**8175 jobs**); the axiom oracle printed, for all **110** registered statements (the 74-entry historical C4 phase-3c cohort plus 36 C5 registrations), exactly

`[propext, Classical.choice, Quot.sound].`

One proof-engineering disclosure: the division-free identity inside the tangency-exclusion proof (`crossing_no_critical_contact`) is closed by `linear_combination` with cofactor $K = -2c^2(2\nu + 1)$; the cofactor was found by polynomial division (SymPy `div`, exact, remainder 0) against the `ring_nf`-normalized goal, and is *replayed entirely inside Lean’s ring normalizer* — the computer-algebra system plays no trusted role. The full run history of this development — seven green build transcripts and the failed-attempt record, each committed the day it was produced with its diagnosis in the commit message; the certified companion script-and-transcript pair; and eight charter amendments recording every external audit round, every score checkpoint, and every method deviation (including the companion’s anchor deviation and depth correction, both registered *before* the certified run) — is committed under `scripts/` and `docs/C5-CHARTER.md`; per the repository’s audit protocol it is archival material and plays no role in the mathematics above.

Headline-theorem-to-artifact map (all in module `AmosClosure`; `AmosClosure/Oracle.lean` enumerates the audited declarations, whereas the certified row refers to numerical evidence rather than to a Lean oracle output; the historical 74-entry cohort is mapped in [11, 12, 13]):

Result	Lean name	File	Status
family, positivity, Eq. (2)	amosFamily, amosFamily_calibration	AmosFamily.lean	exact
Eq. (3) + trichotomy	riccatiQReal_amosFamily etc.	AmosFamily.lean	exact
sufficiency half	amosFamily_upper_of_le_half	AmosFamily.lean	exact
ratio recurrence	besselIReal_ratio_recurrence	AmosLowerReal.lean	exact
lower bound L_ν	besselLowerReal_holds	AmosLowerReal.lean	exact
witness $x_0(c)$	amosFamily_lt_amosLower_witness	AmosLowerReal.lean	exact
Thm. 3.1	amosFamily_uniform_upper_iff	AmosLowerReal.lean	exact
Thm. 3.2	amosFamily_lower_of_one_le	AmosCrossing.lean	exact
Thm. 3.5	besselIReal_turan	AmosCrossing.lean	exact
window (§4.4)	amosFamily_crossing_exists	AmosCrossing.lean	exact
Eq. (4), threshold	crossing_hasDerivAt, crossing_threshold	AmosCrossing.lean	exact
uniqueness, orientation	crossing_unique, crossing_orientation_*	AmosCrossing.lean	exact
scale law	crossing_scale	AmosCrossing.lean	exact
Thm. 3.4	crossing_no_critical_contact	AmosTangency.lean	exact
Thm. 3.3 package	amosFamily_global_crossing	AmosTangency.lean	exact
§6 grid	c5_crossing_arb.py + transcript	scripts/	certified

From a clean clone at the release revision (on Windows, clone with `core.autocrlf=false` — LF bytes reproduce the companion script’s runtime self-hash — and `core.longpaths=true`):

```
lake exe cache get
lake build AmosClosure
lake env lean AmosClosure/Oracle.lean
python scripts/c5_crossing_arb.py
tectonic -X compile papers/c5-crossing/c5_crossing.tex
```

The historical manuscript audited here is the released c5-v1.0.1 artifact. Its annotated tag object is 71097284551d65d989ab7bf024a7c179ee063866, which points to commit 3eb5cf9293d2b8cc6ece17f5e37d422726b9d27d. The original PDF SHA-256 is 95eed84d8c6f85b03bb92ba8f3fe29e3e7664aafb06c4d0c9998e7f5840ec2f0. Release records are in RELEASE-MANIFEST.md and RELEASE-NOTES.md in this paper’s directory; the audit registry is docs/C5-CHARTER.md. Compare historical files with `git show <commit>:path`. This corrected source is a subsequent, separate candidate; it is not the PDF identified by that hash and does not inherit an assessment of those original bytes.

9 Conclusion

For the diagonal Amos-type family at real order, the pinned formal development now contains the parameter classification — uniform upper bound iff $c \leq \frac{1}{2}$, uniform lower bound for $c \geq 1$, with explicit witnesses — and, in the intermediate regime $\frac{1}{2} < c < 1$, a global quantitative crossing theorem: exactly one crossing, transversal, strictly above an explicit threshold, with bilateral orientation, a constructive window, a two-sided scale law, and degenerate contact excluded by an exact curvature identity. Limitations: the in-core series definition only, and $\nu \geq 0$ (Section 5); the family, the classification as mathematics, and the crossing phenomenon are prior art [4] — the quantitative crossing package is the candidate-new content, stated with the priority prudence of Section 5. Natural next steps: the analogous classification for the second-kind ratio $K_{\nu+1}/K_\nu$ (where [6] suggests the tangency structure differs), asymptotics of $x_*(\nu, c)$ beyond the scale law, and upstreaming the modified-Bessel API.

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